

EP05 — Why do computers only understand 0s and 1s?

ENGLISH

01 Title

Why do computers only understand **0s and 1s**? Every photo, every song, every video on your phone — secretly, it's all just two digits. Today, let's find out why.

[CUE] Hold on the title ~2s, calm confident tone, then go.

02 The Hook

Look at your phone. Every photo, every song, every video, every WhatsApp message — **all** of it, deep down, is just zeros and ones. And here's the weird part: we humans count with **ten** digits, zero to nine. So why does the most powerful machine ever built use only **two**? Stick around — in five minutes this whole mystery disappears.

[CUE] Energy up — this is the hook. Hold up the phone on "every photo, every song".

03 Today's question

So here's today's one question: why does a computer use just **two** symbols — and how on earth can two symbols store **everything**? Let's find out.

[CUE] Slow down. Let the question land before moving on.

04 Idea 1 — Switches

Idea number one: a computer is built from billions of microscopic **switches**. Each switch is called a **transistor**. And a switch can only be in two positions — **off**, or **on**. That's it. We just give those two positions names: off we call **zero**, on we call **one**. That's the computer's entire alphabet — two letters.

[CUE] Mime flicking a light switch on "off... on". Reveal 0 then 1.

05 Idea 2 — Two = reliable

Now the big question — why **two**? Why not ten, like us? Here's the key insight. Two states are incredibly easy to **tell apart** — just like a light switch is clearly on or clearly off, which inside the chip means a high voltage or a low voltage. But imagine ten different voltage levels squeezed in. Was that a six or a seven? Tiny bit of noise and you'd make a **mistake**. So two states means simple, and simple means **reliable**, and reliable means fast. That's the whole reason.

[CUE] Land hard on "reliable" — this is the most important idea of the video.

06 Idea 3 — Combine bits

"But two options is so little!" — right? One switch on its own can only say zero or one. The trick is, you **line them up**. Each zero or one is called a **bit**. Put four bits in a row — like one-zero-one-zero — and reading them by place value, that's eight plus two, which equals **ten**. Stack eight bits together and you get a **byte** — that's two hundred and fifty-six different combinations. Add enough bits, and you can represent **any** number at all.

[CUE] Reveal the 8-4-2-1 columns, then the "= 10" result. Read it slowly.

07 Idea 4 — Everything becomes numbers

Okay — but a photo isn't a number, a song isn't a number. So here's the final move. **Everything** first gets turned into numbers. Letters get a number code — that's **ASCII** or **Unicode**. Colours become numbers — a red, green and blue value for each dot. Sound becomes numbers — thousands of tiny samples per second. And once it's all numbers, **every** number becomes binary. So one beautifully simple system stores absolutely everything.

[CUE] Reveal letters → colours → sound rows one at a time as you list them.

08 Why it matters

And this is the beautiful payoff. This is exactly why **"digital"** is so reliable and so cheap. Two clean states — on or off — is why your computer almost **never** makes a mistake reading data. And because each switch is so simple, you can shrink it down and pack billions of them onto a single chip — tiny, cheap, and fast. The entire digital world stands on that one idea.

[CUE] Confident, slightly awed tone — this is the "wow, that's elegant" beat.

09 Common myth

Which clears up a big myth. People think binary is some weird, mysterious **secret code** that only computers can understand. It really isn't. It's just **counting with two symbols** instead of ten — and the only reason it's two is that the hardware physically has two states: on-off switches. There's nothing magical about it at all.

[CUE] Reveal the green "Reality" card after a short beat on the myth.

10 Recap

Quick recap. One: a computer is just billions of **on-off switches** — so its whole alphabet is zero and one. Two: two states are **reliable**, and combining enough of them can represent **any** data — a number, a letter, a photo, a song.

[CUE] Crisp and clear — these are the two lines people should remember.

11 CTA

And that's it — zeros and ones are just switches you can **trust**. If this finally made binary click for you, subscribe to Katixo KhojLab. Next question: how does a computer actually store a **photo** using all these bits? I'll show you. See you there.

[CUE] Warm, friendly. Smile in your voice on the subscribe line.
